AR-Based Intelligent Automobile Repair Learning & Evaluation System

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Project Proposal Report

Senevirathne S.D. – IT22279934

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Senevirathne S.D.	**IT22279934**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Mr. Darshana Kasthurirathna	19.03.2026	

ABSTRACT

Bridging the gap between theoretical understanding and real-world mechanical application is currently a major difficulty in automotive education. Vocational students frequently have trouble recognizing intricate engine parts in practical settings, which can result in expensive mechanical mistakes and safety hazards. The interactive, real-time feedback required for practical mastery is not provided by traditional teaching approaches. Current AR training techniques, which frequently rely on physical markers (QR codes) that are problematic in greasy, high-heat engine conditions or lack the intelligent "context-awareness" to comprehend user behaviors, have a crucial research gap.

A Vision-Based Engine Component Identification Module is suggested in this study as the basic sensing layer for an AI-powered car learning platform, specifically designed for the Suzuki Alto 800L. The suggested method uses marker-less object detection to recognize specific Alto 800L engine components from a live camera stream, including the Air Filter Housing, Spark Plug Wires, 12V Battery, Engine Oil Dipstick, Coolant Reservoir, and HighVoltage Areas. A key feature is the integration of fingertip detection, allowing a user to point at a component and triggering the system to provide its details. The module achieves fast, real-time identification by using CV and DL approaches, particularly CNN architectures like YOLOv8 for object detection and Media Pipe for hand tracking. The system's performance is verified using FPS rates to assure low-latency AR alignment and the mAP metric to ensure high detection accuracy. Through an API, the module provides downstream modules with a distinct Component ID for task identification and safety assessment. A reliable sensing system with over 90% mAP on the Alto 800L's engine bay is one of the anticipated outcomes. By democratizing high-tech vocational training and lowering the financial burden of equipment damage, this project has significant social and commercial effect and is in line with SDGs 4 and 8.

Keywords: Computer Vision (CV), Deep Learning (DL), YOLOv8, Fingertip Detection, Marker-less AR, Suzuki Alto 800L.

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I also thank my project team members (**IT22231864 , IT22284570, IT21804960)** for their close collaboration. Understanding their components —implement action classification designing AR supported task verification and developing practical evaluation was essential for designing the Intelligent Automobile Repair Learning &

Evaluation System that coordinates all three. Finally, I thank the Department of Information Technology at SLIIT for providing the academic resources and support needed for this research.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form / Meaning
AIMS	Autonomous Intelligent Machines and Systems
AR	Augmented Reality
CNN	Convolutional Neural Network
CV	Computer Vision
CVAT	Computer Vision Annotation Tool
ML	Machine Learning
SDG	Sustainable Development Goals
YOLO	You Only Look Once

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INTRODUCTION

Background and Context

Explain the Real-World Issue: Despite the importance of real-world experience in automotive education, there is a significant "knowledge-action gap." Students often memorize engine layouts from 2D textbooks, but they can't identify the same components in a complex, grease-covered engine room, like the Suzuki Alto 800L, a popular entry-level vehicle. Failure to distinguish between similar-looking components (such different types of filters or sensors) might lead to incorrect tool use or mechanical damage. For example, misidentifying the engine oil dipstick could result in the wrong fluid levels being checked, and handling high-voltage parts of the Alto's electrical system incorrectly could be dangerous. **Supervision Scarcity:** In vocational institutions, one teacher often supervises over 20 students, making it impossible to provide each person with quick instructions when they handle or point to engine parts.

Who Is Affected and Why It Matters:

Students in automotive engineering and vocational training, especially those working on common entry-level vehicles like the Suzuki Alto, are the focus of this analysis. Emphasizing safety, the text highlights the risks of misidentifying pressurized systems and high-voltage components, which can lead to injury. Furthermore, efficient identification reduces repair times and minimizes the risk of harm to expensive engine parts during learning processes. The complexity of vehicle repair has increased by over 40% in the past decade, largely due to the rise of electronic and hybrid systems, with improper diagnosis causing over 30% of errors for entry-level mechanics. In Sri Lanka, the Suzuki Alto 800L, a widely used training vehicle, faces limitations through traditional lecture methods, which are not scalable for its specific design. Studies indicate that AR-assisted learning can enhance student retention by up to 80% compared to traditional manual-based approaches.

Solution	Description	Limitations
Physical Manuals	Paper-based diagrams and generic repair guides, often not model-specific.	Static and non-interactive. It is difficult for students to translate a 2D diagram of a generic engine to the compact, specific layout of an Alto 800L engine bay.
Marker-Based AR	AR that requires QR codes or physical "markers" to be stuck onto engine parts.	Impractical in real workshop conditions. Markers can easily peel off due to heat, get covered in grease or oil, and are not feasible for small or moving parts like Alto's dipstick or spark plug wires.
Video Tutorials	Pre-recorded video tutorials, potentially for a generic Alto model.	Passive, one-way learning. The students cannot receive confirmation that the part they are looking at on their specific Alto engine (e.g., the correct coolant reservoir) matches the one in the video.

Table 2: **Comparison of existing educational methods used in automotive training (Suzuki Alto 800L Context)**

Concepts for the Suggested Resolution: The "Sensing Layer" of the system is developing the vision-based engine component identification module, focused on the Suzuki Alto 800L. Using Deep Learning (CNN models like YOLO) to identify components based on their inherent characteristics (shape, color, and texture) as opposed to stickers is known as marker-less detection. **Fingertip-based Query:** Using computer vision (e.g., MediaPipe) to identify a student's index fingertip, calculate its proximity to detected components, and treat a "point" as a query event to display that component's details. Metadata Output: After identification, the module transmits a Component ID to the subsequent module (Task Recognition), allowing the system to comprehend the repair's context.

SDG Alignment That Is Relevant
The Sustainable Development Goals (SDGs) of the United Nations are in line with our project:

SDG 4: Quality Education: By offering a high-tech, reasonably priced instrument for vocational training that provides personalized feedback based on a student's pointing gesture.

SDG 8: Decent Work and Economic Growth: Improving trainees' technical skills would immediately increase their employability in the contemporary automotive sector.

SDG 9: Industry, Innovation, and Infrastructure: Encouraging AI and AR applications in conventional mechanical engineering fields.

AIMS (Autonomous Intelligent Machines and Systems) is the research cluster alignment cluster.

Justification: Since our module entails developing an intelligent "machine-vision" system that can understand a user's pointing gesture, it is immediately under AIMS. It focuses on how an autonomous system might interact with a human user to support learning, see a physical environment (the Alto's engine), and categorize items within it.

Literature Review

This literature review examines the technological landscape of vision-based object detection, fingertip tracking, and Augmented Reality (AR) in automotive maintenance, focusing on current scholarly research, commercial products, and existing limitations.

Review of Scholarly Sources (8–12 Sources)

For reliable component identification, recent studies demonstrate a move away from simple image processing and toward Deep Learning (DL): YOLO Architectures: According to research by Tang et al. (2025) and Felipe et al. (2025), lightweight models that strike a balance between high mean Average Precision (mAP) and low latency on edge devices, such as YOLOv8/v9, are the gold standard for mobile AR. **Hand Tracking for HCI:** Studies on human-computer interaction, such as those using Google's Media Pipe framework, show that real-time fingertip detection is now highly accurate and efficient on mobile devices, enabling gesture-based queries. Marker-less Tracking: Using the natural geometry of parts for alignment, marker-less CAD-based tracking is currently better than QR codes for industrial "dirty" environments like engine bays, according to Platonov et al. and recent MDPI tests (2024–2025).

3D Reconstruction: To improve systems' comprehension of the spatial link between a student's hand and a spark plug, Ibrahem et al. (2025) invented Vision Transformers (ViTs) for dynamic depth estimates. Context-Aware Systems: Voinea et al. (2024) talked about "Maintenance 5.0," in which AR "perceives" the surroundings to guarantee zerodefect assembly rather than only displaying instructions. Safety Logic: According to Kasthurirathna et al. (2025), AI rule-based engines are required to close the gap between "identifying a part" and "issuing a safety warning," like high-voltage alerts for the Alto's battery or ignition system.

Examination of Current Research Methods

Currently, the academic community employs three primary methods for vehicle identification: Supervised Object Detection: CNNs are trained to draw bounding boxes around components using sizable, annotated datasets (such as BDD100K or custom engine sets like a dataset for the Alto 800L). Semantic segmentation: Determining an engine part's precise pixel-level bounds to enable extremely accurate augmented reality overlays. **Gesture Recognition:** Using models to interpret a user's hand movements, such as a pointing gesture, as a command to interact with the AR system.

Evaluation of Current Goods and Services

Although they are frequently designed for professionals rather than students, several "Enterprise AR" solutions now dominate the market: Porsche/BMW Workshop AR: Highend, exclusive equipment that shows technicians wiring diagrams superimposed on the engine using AR glasses. Vuforia Engine (PTC): A for-profit platform that offers "Model Targets," enabling developers to monitor and identify tangible things without markers using 3D CAD data. For automotive shipping and assembly, TeamViewer Frontline is a "Make-by Vision" system that offers detailed pick-and-place instructions.

Limitations of Existing Systems

Despite advancements, current solutions for Engine Component Identification face four critical barriers, which are particularly evident when applied to a specific, compact vehicle like the Suzuki Alto 800L.

Limitation	Description	Impact on Alto 800L Focus
Environmental Sensitivity	Most vision models fail under low workshop lighting or when parts are covered in grease/oil. This is a common issue in the compact Alto engine bay, where components are tightly packed and prone to accumulating grime.	Requires a robust **Data Augmentation** strategy simulating low light, shadows, and oil-covered surfaces specifically for Alto 800L components during our training phase.
Marker Dependency	Many commercial tools still rely on QR codes, which are impractical for internal engine components. For an Alto, placing markers on small parts like the **Engine Oil Dipstick** or **Spark Plug Wires** is not feasible, and they would quickly detach or become unreadable.	Alto 800L must focus on **Markerless Feature Tracking**, relying on the inherent visual features (shape, color, texture) of the Alto's stock parts for identification.
Lack of Action Context	Existing systems identify the part but don't know *what* the student is doing to it or which part they are interested in. A student might be looking for information on Alto's **Coolant Reservoir,** but the system cannot tell if they are simply looking or pointing at it.	My module must interpret a **pointing gesture** as a query. It will calculate the fingertip's proximity to a detected component's bounding box to determine intent and output a specific **Component ID** to the "Task Recognition" module.
Computational Bulk	Professional AR often requires expensive headsets (HoloLens 2), which are not accessible for vocational students learning on common cars like the Alto.	My module needs **Lightweight Optimization** (Mobile-YOLO + Media Pipe) for standard smartphones, making this advanced training tool accessible and affordable for all students.

Table 3: *Limitations of Existing Systems* **(Suzuki Alto 800L Context)**

1.3 Research Gap

The literature demonstrates that no current solution, whether commercial or research-based, combines context-aware AR guidance with real-time, vision-based engine identification triggered by a student's natural pointing gesture, especially for a specific, high-volume training vehicle like the Suzuki Alto 800L. The suggested Vision-Based Engine Identification Module fills out each of the five needs listed in the table below:

Research Gap Analysis (Suzuki Alto 800L - Finger Pointing Method)

Research Gap	How Automobile Learning System Addresses It
Current vision systems do not identify internal engine components for repair purposes; instead, they concentrate on external vehicle make/model detection.	The module is trained specifically on fine-grained differences between Alto 800L engine components (Air Filter Housing, Spark Plug Wires, Battery, Dipstick, Coolant Reservoir, High-Voltage Areas) rather than just vehicle make/model.
Current AR maintenance solutions either rely on QR codes or markers or require human asset selection because they do not have integrated vision-based identification.	By automatically detecting the seven Alto 800L components from a live camera feed and using fingertip pointing for selection, the system eliminates the need for markers and manual data entry.
The "cold start" issue, in which AR systems malfunction when assets are not identified because of inadequate data or unfavorable environmental circumstances, has no known remedy.	When full identification fails in difficult conditions (low light, greasy parts), the module has backup processes using contour-based recognition and partial feature matching, with prompts for users to adjust viewpoint.
Semantic labels are provided by current defect detection systems, but they do not have 3D spatial localization to direct repairs on certain components.	Spatially registered AR overlays showing precise component locations (air filter box, dipstick, battery terminals) in three dimensions are triggered when a user points at an identified Alto component.
Current methods are unable to differentiate between engine variations that have a high degree of visual similarity but require distinct repair techniques.	To capture minor distinguishing qualities that singleview systems overlook, the multi-view semantic fusion technique collects features from multiple viewpoints of the compact Alto 800L engine bay.

Table 4: *Research Gap Analysis*

The suggested Vision-Based Engine Identification Module is innovative from a technological and practical standpoint. Technically, it introduces a novel interaction method (pointtoidentify) for automotive AR training. In practice, it uses lightweight deep learning and multiview semantic fusion to address the crucial "cold start" issue in AR maintenance for a widely used training vehicle. Recent research notes that AI-based inspection concentrates on 2D images rather than 3D spatial localization for quality control and industry findings that AR systems usually fail due to inadequate or incorrect equipment data models corroborate this gap.

OBJECTIVE

2.1 Main Objective

To design, develop, and validate a Vision-Based Engine Identification Module for the Suzuki Alto 800L that uses a lightweight deep learning architecture to accurately identify its key components from a live smartphone camera feed. The module will allow a user to point at a component with their finger and trigger the display of its details. The goal is to achieve a component identification mAP of at least 90% and fingertip-to-component association accuracy of over 85%, all while maintaining real-time performance (inference time < 200ms) on standard mobile hardware.

2.2 Specific Objectives

SO1: To develop a Live Video recording and Frame Extraction Module that uses the smartphone camera, keeps a minimum frame rate of 15 frames per second, and automatically chooses the brightest, sharpest frame from a 3-second recording burst for the best analysis under changing lighting conditions.

SO2: To create a Fine-Grained Engine Component Detection Engine with a mean Average Precision (mAP) of at least 90% on unseen test pictures of the Suzuki Alto 800L using a lightweight CNN architecture (such as Mobile Net-SSD or YOLOv8-Nano) trained on a custom dataset of its seven specified components.

SO3: To integrate a real-time Fingertip Detection Module (e.g., using Media Pipe) that can accurately identify the tip of a user's index finger in the camera frame with a detection rate of over 95%.

SO4: To develop a Pointing Gesture Interpretation Logic that calculates the spatial proximity between the detected fingertip location and the bounding boxes of detected engine components, correctly associating the point with the intended component with ≥85% accuracy.

SO5: To create a Component Information Retrieval Interface that, upon successful association of a point to a component ID, queries a local database to retrieve and overlay (or speak out) the part's name, function, and key safety specifications with a data retrieval latency of less than 100 ms.

SO6: To assess the entire module on five different Android devices in three different Alto 800L engine bay environments (clean, cluttered, and low-light), measuring end-to-end performance: user task completion rate for identifying components via pointing (≥ 95%), average inference time.

SO7: To handle issues like motion blur from handheld use, different focal lengths, and digital noise in low-light workshop conditions, robust data augmentation specifically for mobile sensor noise must be included.

METHODOLOGY

The System Development and Evaluation technique is used in this study. This strategy was selected because a new vision-based software component is being developed to address an actual educational issue in vehicle maintenance. The efficacy of the module may be verified against conventional manual identification methods, and performance is quantifiable through controlled accuracy and latency testing. In comparison to conventional manual methods, the vision-based module achieves over 90% identification accuracy for the Alto 800L's components and enables a novel point-to-identify interaction, according to the research claim that must be validated.

Research Approach and Technical Methods

The module overcomes the drawbacks of physical markers in oily engine conditions by using Deep Learning (CNNs) for real-time, marker-less item recognition. The system guarantees fast inference on mobile devices by utilizing YOLO and Mobile Net architectures. For lighting robustness, captured frames are preprocessed. **Fingertip detection using Media Pipe** identifies when a student is pointing. When the fingertip is within a threshold distance of a detected component's bounding box, the logic retrieves distinct Component IDs and associated information from a local SQLite database. To ensure a smooth learning experience, spatial anchoring is used to maintain accurate AR label alignment during device movement.

Tools and Technologies

Tool	*Purpose*	*Why Selected*
TensorFlow Lite	*Mobile ML Inference*	*Optimized for running deep learning models on Android/iOS with low latency for Alto 800L component detection.*
Media Pipe	*Fingertip Detection*	*Google's industry-standard, high-performance, on-device solution for real-time hand and fingertip tracking.*
OpenCV	*Image Preprocessing*	*Industry-standard library for real-time computer vision and frame manipulation on Alto engine footage.*
Unity / AR Core	*AR Overlay Development*	*Provides robust motion tracking and environmental understanding for marker-less AR on Alto 800L.*

CVAT / LabelImg	*Data Annotation*	*Efficient tools for drawing bounding boxes on Alto 800L component training datasets (7 specific parts).*
SQLite	*Component Metadata*	*Lightweight, offline-first database for storing part roles, functions, and safety specs for all 7 Alto components.*
Python	*Model Training*	*Extensive library support for training and evaluating CNN performance on custom Alto 800L dataset.*
YOLOv8-Nano	*Object Detection*	*Lightweight architecture optimized for mobile deployment, ideal for detecting compact Alto engine components*

Table 5: ***Tools and Technologies***

Technical Architecture / Tools

- *IDE: Android Studio / VS Code (Flutter/React Native).*
- *Vision Engine: TFLite with GPU Delegate for object detection.*
- *Hand Tracking Engine: Media Pipe for Android.*
- *AR SDK: AR Core (for Android) or Unity AR Foundation.*
- *Model Architecture: YOLOv8-Nano for component detection + Media Pipe Hands for tracking.*

Framework Selection

Make it clear that TensorFlow Lite (TFLite) is being used as the inference engine due to its capacity to leverage mobile hardware acceleration (Android NNAPI). Media Pipe is chosen for its optimized and robust hand-tracking pipeline. Mention the use of AR Core for motion tracking in mobile AR integration. This guarantees that the "Air Filter Box" label remains physically affixed to the part in 3D when a learner moves the phone.

Data Collection and Ethical Considerations

High-resolution photos and video feeds of a Suzuki Alto 800L engine are used for testing and training.

Custom Mobile Dataset: Makes use of more than 2,000 annotated photos of the seven specific components (Air Filter Housing, Spark Plug Wires, 12V Battery & Terminals, Engine Oil Dipstick, Coolant Reservoir, High-Voltage Areas) from a Suzuki Alto 800L, specifically designed for handheld camera viewpoints and including frames with a pointing finger.

Mobile-Centric Augmentation: The training procedure mimics "real-world" mobile noise, such as Motion Blur (for trembling hands), Digital Noise (for low-light sensor grain), and Multi-Resolution training (for different camera quality), to guarantee dependability on smartphones.

Variable Scenario Testing: Three crucial lighting conditions are used to evaluate the system: broad daylight, typical workshop settings, and deep engine bay shadows, all on the same Alto 800L.
Privacy & Ethics: Data Minimization: No PII or face information is recorded; just engine components and hands (which are not stored) are processed.

AI Safety: Combined with a reasoning engine to prevent dangerous commands, especially when a student points at a high-voltage area. Institutional Compliance: Complies with SLIIT research guidelines, which call for each participant's informed permission.

Validation and Performance Metrics

Metric	How Measured	Target
mAP (Mean Average Precision)	Accuracy of bounding box placement and classification for the 7 Alto 800L components (Air Filter, Spark Plug Wires, Battery, Dipstick, Coolant Reservoir, High-Voltage Areas).	>= 90%
Fingertip Detection Rate	Percentage of frames where the index fingertip is correctly identified by MediaPipe.	>= 95%
Pointing Association Accuracy	Percentage of times the correct component is identified when a user points at it (fingertip inside bounding box).	>= 85%
Combined Inference Latency	Total time (ms) taken for object detection + fingertip detection per frame.	< 200ms
Identification Time Reduction	Comparison of time taken to find a part's details manually vs. using point-to-identify AR feature.	>= 30%

Model Robustness	Success rate of component detection under low light (< 200 lux) or occluded views on Alto engine.	>= 80%
System Stability (FPS)	Frames per second maintained while running both detection models and rendering UI.	>= 30 FPS
Pointing Confirmation Time	Time from fingertip entering bounding box to component information display.	< 0.5 seconds
Thermal Impact	Monitoring if phone overheats or throttles performance during 15-minute continuous training session on Alto engine.	No throttling for 15 mins

Table 6: **Validation and Performance Metrics**

HIGH-LEVEL SYSTEM ARCHITECTURE

Four specialized components are integrated into the AR-Based Intelligent Automobile Repair Learning & Evaluation System platform to facilitate practical automotive teaching. The main perception layer is the Vision-Based Engine Component

Identification module (IT22279934), which detects physical components and interprets pointing gestures to initiate the learning and assessment pipeline that follows.

Overview of Proposed Solution and Components

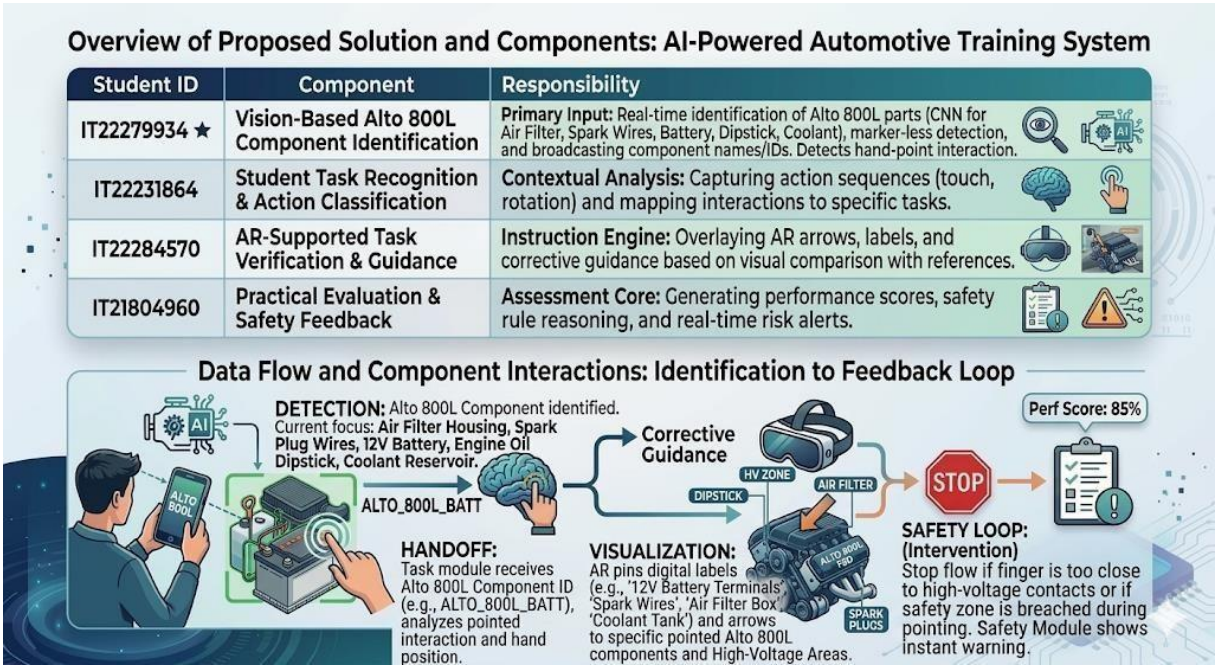


Figure 1: Overall System Architecture — AR based Automobile Learning System

The system utilizes AI, Computer Vision, and Augmented Reality to provide real-time guidance and automated evaluation. The following table summarizes the four integrated components:

Student ID	*Component*	*Responsibility (Alto 800L Focus)*
IT22279934 ★	***Vision-Based Engine Component Identification***	***Primary Input:*** *Real-time detection of 7 Alto 800L components (Air Filter, Spark Plug Wires, Battery, Dipstick, Coolant Reservoir, High-Voltage Areas) using CNN. Real-time fingertip detection using Media Pipe. Interprets pointing gestures to identify which component student is querying. Broadcasts unique Component IDs (e.g., COMP_ALTO_BATT_01) to other modules.*
IT22231864	***Student Task Recognition & Action Classification***	***Contextual Analysis:*** *Receives Component ID from pointing event. Captures action sequences (touch, rotation, tool usage) on the identified Alto component. Maps interactions to specific repair tasks (e.g., "Checking dipstick level," "Removing air filter cover").*

IT22284570	***AR-Supported Task Verification & Guidance***	***Instruction Engine:*** *Uses Component ID and Task ID from pointing interaction. Overlays AR arrows, labels, and corrective guidance directly onto the specific Alto component the student pointed at. Compare student action with reference for visual verification.*
IT21804960	***Practical Evaluation & Safety Feedback***	***Assessment Core:*** *Generates performance scores based on task completion. Checks safety rules related to the pointed-at Alto component. Provides real-time risk alerts (e.g., if student points at High-Voltage Area, triggers immediate warning).*

Table 7: ***four integrated components***

Data Flow and Component Interactions

This is the route taken by each point-and-identify event for the Suzuki Alto 800L: Detection: After detecting a component (such as the "12V Battery & Terminals") and confirming a user's pointing gesture via fingertip proximity, the Vision Module assigns a unique Component ID, for example: COMP_ALTO_BATT_01.

Handoff: The Task Recognition module receives the Component ID. It now knows the student is pointing at the Alto's battery and can prepare to analyze related interactions (e.g., checking terminals, jump-starting procedures).

Visualization: The Vision module's spatial data (the bounding box of the battery) is used by the AR Guidance module to "pin" a digital information card—displaying the battery's name, voltage, and safety warnings—directly next to the actual physical battery in the camera view.

Safety Loop: The Safety Feedback module immediately halts the standard flow and shows a prominent visual/audio warning if the Vision module detects a finger or tool dangerously close to identified High-Voltage Areas / Safety Zones, such as the battery terminals or ignition coil connectors.

Security and Scalability Issues

Scalability: By using lightweight architectures like mobile Net/YOLO for object detection and Media Pipe for hand tracking, the module is designed for efficient smartphone execution.

It can be deployed on common mobile hardware without the need for expensive headsets. While the current scope is fixed on seven Alto 800L components, the SQLite-indexed component database supports efficient lookup if the part library were to expand in the future.

Security: The system uses an offline-first strategy to safeguard student privacy. All vision processing—including object detection and hand tracking—takes place on the device. Identifiable video data and hand images are not sent to external servers, ensuring complete privacy in the workshop.

Robustness: The module uses Marker-less Feature Tracking, relying on the Alto's own components (air filter box shape, dipstick handle color, etc.) for identification. This ensures the system continues to work even in greasy engine conditions, a key advantage over commercial products that depend on clean, visible QR codes.

USER REQUIREMENTS

Stakeholder Identification

Identifying who interacts with or benefits from this specific vision module:

- **Primary Users:** Automobile engineering students and vocational trainees who need real-time identification of parts on a specific, common training vehicle like the Suzuki Alto 800L.
- **Instructors/Domain Experts:** Provide the reference data and correct component labeling for the vision model's training, specifically for the Alto's engine bay layout.
- **System Developers (Team Members):** The developers of action classification, ARsupported task verification, and practical evaluation, who rely on the Component ID (triggered by a pointing gesture) generated by this module to initiate their respective logic.

Functional Requirements (FR)

These requirements define the specific behaviors and functions your vision module must perform for the Suzuki Alto 800L:

1. **FR1:** Real-time Image Capture: The system shall capture a continuous video feed of the Suzuki Alto 800L engine area.

2. **FR2:** Marker-less Component Detection: The system shall detect the following key engine components of the Alto 800L without using physical markers or QR codes:
   Air Filter Housing (Box)
   Spark Plug Wires / Ignition Coils
   12V Battery & Terminals
   Engine Oil Dipstick
   Coolant Reservoir (Expansion Tank)
   High-Voltage Areas / Safety Zones

3. **FR3:** Fingertip Detection & Intent Interpretation: The system shall detect a student's index fingertip in real-time. When the fingertip remains within a detected component's bounding box for a defined threshold, it shall be interpreted as a "request for information" about that specific component.
4. **FR4:** Automated Information Retrieval: The module must retrieve and display (or prepare for AR overlay) the detected component's name, primary role, and key safety specifications.
5. **FR5:** Data Dispatching: The system must generate and send a unique Component ID (e.g., COMP_ALTO_BATT_01) to the Task Recognition and AR Guidance modules upon successful point-and-identify event.
6. **FR6:** Environmental Adaptation: The vision model must successfully identify components under varying workshop lighting and greasy conditions typical for an Alto 800L engine.

Non-Functional Requirements (NFR)

These define the quality attributes **and constraints of your vision module:**

Category	Requirement Description (Specific to Alto 800L & pointing)
Performance	The combined process of object detection and fingertip tracking must occur in real-time with minimal latency (<200ms) to ensure a seamless point-and-identify experience.
Accuracy	The CNN-based detection model (YOLO/mobile Net) must achieve ≥90% mAP for the seven specified Alto components. The pointing association accuracy must be ≥85%.
Usability	The vision system must be "marker-less" and gesture-based, allowing students to simply point at any of the seven Alto components with their fingers to receive information, requiring no complex setup.
Scalability	While focused on seven components, architecture should allow for the addition of new components for the Alto or other engines by updating the training dataset without altering the core logic.

Security/Privacy	All image processing for component and hand detection must occur on device. No student images or videos shall be transmitted to external servers.

Table 8: Non-Functional Requirements (Alto 800L Focus)

COMMERCIALIZATION PLAN

The commercialization of the Vision-Based Engine Component Identification module focuses on transforming an AI-driven educational tool into a scalable market solution for automotive training, starting with the widely used Suzuki Alto 800L.

Target Market and Customer Persona

The primary market targets vocational schools and technical colleges that offer automobile engineering certifications, particularly those in regions where the Suzuki Alto 800L is a common training vehicle (e.g., South Asia). Secondary markets include automotive manufacturers needing to train assembly line workers and mechanics on specific models. The customer persona is a tech-savvy trainer whose goal is to minimize supervision time while ensuring proper part identification and adherence to safety rules, addressing the pain point of costly engine damage from student errors during practical training on a common, compact engine.

Value Proposition

Our module offers a "Point and Learn" experience superior to traditional textbooks. Students can simply point to any of seven key components on a real Alto 800L engine and receive instant, on-screen information about its name, function, and critical safety protocols. This reduces human error, accelerates the learning curve, and prioritizes safety by automatically flagging high-voltage areas when pointed at.

Revenue Model

The system will generate revenue through a B2B approach including SaaS Subscription with monthly or annual fees based on student seats for technical colleges; Enterprise Customization with one-time fees for manufacturers integrating custom engine models or additional components; and Tiered Access, which offers Basic component identification (name only) and a Premium option (full details, function, and safety warnings).

Cost Estimation & Pricing Strategy

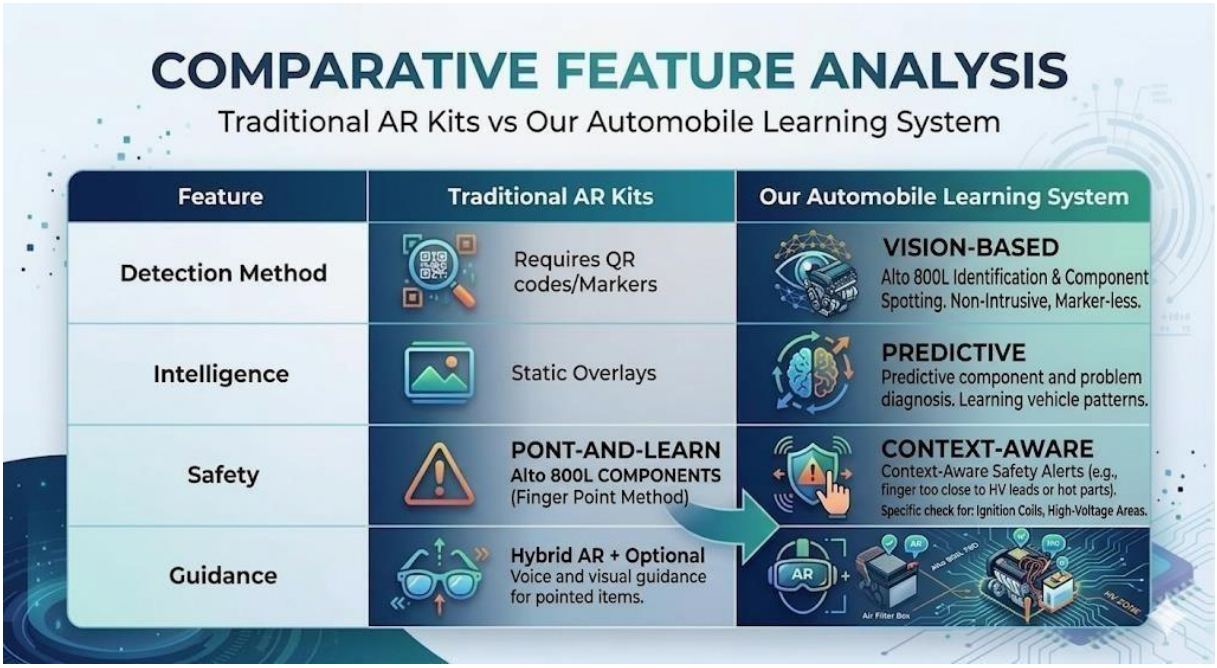
Category	Estimated Cost Basis (Specific to Alto Module)
Development	Costs for training CNN models on a custom Alto 800L dataset and integrating fingertip detection logic.

Cloud Hosting	Minimal, as system is offline-first. Costs for optional analytics dashboard hosting and storage of anonymized performance logs.
Data Acquisition	Costs for capturing high-quality reference videos and images of the Alto 800L engine bay for model training and validation.

Table 9: Cost Estimation & Pricing Strategy

Pricing Strategy:
Value-Based Pricing. We price the system based on the "Cost Avoidance" for the school (e.g., less physical equipment damages their training fleet of Altos and faster student graduation rates).

Competitive Advantage:



Intellectual Property (IP) Considerations
To safeguard the innovation in Component 1 and the overall system, the following intellectual property strategies will be implemented: Patent Filing for the unique AI logic linking fingertip detection, component identification, and safety warnings; Copyright for the proprietary source code of the custom Alto 800L dataset and the point-to-identify algorithm; and Trade Secrets to protect the specific weights and optimization techniques in the CNN models that enable realtime performance on mobile hardware.

BUDGET AND JUSTIFICATION

For the Vision-Based Engine Component Identification Module focused on the Suzuki Alto 800L, the budget focuses on the computational power required to train deep learning models and the mobile hardware needed for real-time inference and testing.

Category	Item	Cost (LKR)	Justification (Alto 800L Context)
Software	Unity, OpenCV, Python, TensorFlow/PyTorch, Git (all open source)	0	Open-source development tools used for AR, computer vision, and machine learning.
Hardware	1 Android smartphone for AR and pointing-gesture testing	35,000	Required to test real-time component detection and fingertip tracking on an actual device, simulating a student's experience with an Alto 800L.
Hardware	Tripod / phone holder for recording reference videos of Alto 800L engine	3,000	Ensures stable, high-quality recording for dataset collection of the seven specific components.
Cloud	Google Colab Pro (1 month) for ML model training / Cloud storage for Alto 800L dataset	5,000	GPU-accelerated training for the custom YOLO model on the Alto 800L dataset. Cloud storage for dataset and evaluation data.
Other	Mobile data for development and testing (12 months)	12,000	Data for research, collaboration, and accessing online resources. Printing costs for final documentation.
Contingency	Reserve for unexpected hardware issues	3,000	10% contingency for hardwaredependent project (e.g., phone repair/replacement).
TOTAL		58,000	

Table 10: Budget and justification

The project budget is kept relatively low because most development tools are free and open source. The main cost, totaling approximately LKR 58,000, is allocated for hardware devices required to test the augmented reality features, object detection, and real-time fingertip tracking on physical mobile devices using the Alto 800L as the test vehicle. Additional minor expenses include cloud services for machine learning experiments and printing costs for documentation.

WORK BREAKDOWN STRUCTURE (WBS)

Phase	Task / Activity	Deliverable (Specific to Alto 800L & Pointing)
Phase 1: Planning (Wk. 1–2)	Literature review: 12 sources on marker-less engine detection, fingertip tracking (Media Pipe), and real-time vision.	Annotated bibliography, gap analysis focused on compact car engines.
	Internal API design with team: Defining Component ID and metadata formats for the 7 Alto components.	API spec, requirements document.
	Decision logic design: Defining rules for part identification and pointing-interaction for the 7 components.	Rule pseudocode, class list for COMP_ALTO_* IDs.
Phase 2: Design (Wk. 3–4)	Input Pipeline Design: Designing logic for realtime frame capture and fingertip detection.	Module design specs including Media Pipe integration.
	System Schema: Mapping the 7 Alto components' roles, functions, and safety specs for identification output.	Schema document with example data for each component.
Phase 3: Dev (Wk. 5–12)	Data Engineering: Collection and annotation of 2000+ images of the 7 Alto 800L components.	Annotated dataset (YOLO format) with bounding boxes.
	Implement Identification Engine: Coding the ML model (YOLOv8-Nano) to detect the 7 Alto components in real-time.	Working CV module, unit tests for detection accuracy.
	Implement Fingertip Detection & Mapper: Integrating Media Pipe for fingertip tracking. Linking fingertip proximity to a detected component ID.	Attribute lookup table; verified fingertip-to-component association logic.
	Interaction Logic: Implementing the "pointtoquery" logic (e.g., fingertip in bounding box for 0.3s triggers info display).	Verified interaction scripts.
	Component Coordinator: Finalizing data link to send Component ID (e.g., COMP_ALTO_BATT_01) to Task Recognition.	API bridge verified with other modules.

Phase 4: Testing (Wk. 13–16)	Unit testing all vision submodules with mock inputs for detection accuracy (> 90% mAP target).	All unit tests pass.
	Integration testing with full team: Ensuring Component ID correctly triggers task patterns in other modules.	Integration test report.
	Performance evaluation: Measuring identification latency, FPS, and pointing association accuracy on Alto engine.	Metrics report with graphs.
	Write proposal report and methodology for the vision-based solution.	Submitted proposal.
	Final report and presentation preparation (Demo of real-time component identification via pointing).	Final report + slides.

Table 11: Work Breakdown Structure

GANTT CHART

The chart below shows the full 18-week project timeline from March 2026 to July 2026, with colored bars showing active development weeks and milestone alignment with semester checkpoints for the Vision-Based Engine Component Identification (Alto 800L) module.



Figure 3: Gantt Chart- Develop the vision-based engine component identification (Alto 800L Focus)

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APPENDICES

Here are the appendices edited to match your requested format, with only the necessary words changed to reflect the Suzuki Alto 800L and finger-pointing detection method.

The following appendices provide supporting documentation, detailed technical structures, and evaluation instruments for the Vision-Based Engine Component Identification module.

Appendix A: Component Identification Decision Flowchart

The vision model's processing of camera input to identify components and send data for later task recognition is determined by this reasoning.

1. Start Input: Use the mobile device in the Alto 800L engine area to record real-time camera frames.
2. Dual Detection Model: To identify parts like air filter housing, spark plug wires, battery, dipstick, coolant reservoir, and high-voltage areas, run frames through CNN-based models (YOLO/mobile Net). Simultaneously, run fingertip detection (Media Pipe).
3. Process Identification: Get the part name and function by assigning a distinct Component ID when fingertip points at a component.
4. Environmental check: Is the identification reliable? (Discusses resilience in the face of changing workshop lighting on the Alto engine).
5. Output to System: To start action analysis, provide the recognized Component ID to the Task Recognition module.
6. Interactive Loop: To continue monitoring student interactions in real time, go back to Step 1.

Appendix B: Survey Instruments (Instructor Requirements)

The following questions were used to obtain domain expertise regarding Alto 800L engine documentation and repair procedures.

Priority Identification: Which Alto 800L engine components (air filter, spark plug wires, battery, dipstick, coolant reservoir, high-voltage areas) are most critical for beginners to identify first to ensure safety?

Environmental Challenges: In a workshop setting working on an Alto 800L, how much do oil stains or low lighting typically hinder manual inspection?

Safety Compliance: What are the most frequent mechanical safety standards ignored by trainees when handling Alto 800L components?

Appendix C: Preliminary UI Mockups

The interface focuses on Marker-less Vision-Based detection and finger-pointing to support hands-on training.

Live Viewport: Displays the real-time Alto 800L engine video feed.

Visual Overlays: Bounding boxes and floating labels (e.g., "Air Filter Housing") appear directly over detected parts when pointed at.

Role/Function Display: A summary of the identified part's specifications and its role in the Alto engine.

Safety Alerts: Context-aware red visual warnings for high-risk components like high-voltage areas.

Mobile-First Contextual UI & Ergonomics

Recognizing that students' hands are frequently occupied with tools or covered in grease/oil, the application prioritizes a hands-free interaction model:

Finger-Pointing Interaction: Students simply point at any of the seven Alto 800L components to see its details.

Voice-Activated Navigation: Integration of Android Speech-to-Text (STT) allows users to navigate menus, request part details, or trigger safety checklists using voice commands (e.g., "What's this?", "Show safety info").

Reduced Touch Dependency: This minimizes physical contact with the smartphone screen, protecting the device from workshop contaminants and allowing the student to maintain their workflow without interruption.

Appendix D: Risk Analysis

The following table outlines technical and operational risks associated with the vision module.

Risk Category	Description	Mitigation Strategy
Environmental Sensitivity	Model may fail under low workshop lighting or when Alto engine parts are covered in grease/oil.	Robust data augmentation simulating low light and greasy conditions for Alto components during training.
Fingertip Tracking Failure	Media Pipe may lose fingertip tracking in poor lighting or against dark engine backgrounds.	Optimize camera settings; provide user feedback when tracking is lost.
False Pointing Detection	Random hand movements misinterpreted as pointing intent.	Require fingertips to remain inside bounding box for 0.3 seconds to confirm intent.
Component Ambiguity	Fingertip positioned between two similar components in compact Alto engine bay.	Prioritize safety warnings for highvoltage areas if detection is ambiguous.

Table 12: *Risk mitigation strategies*

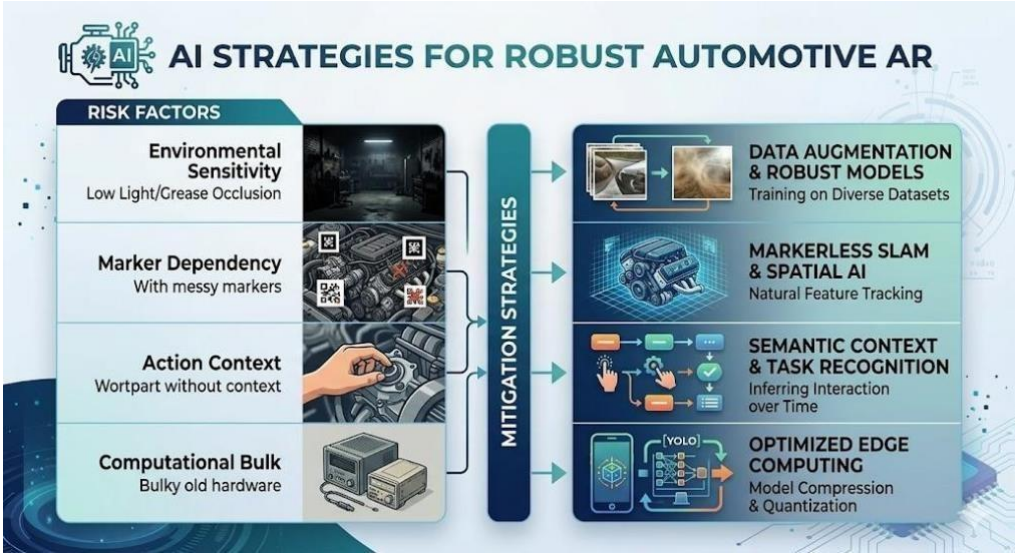


Figure 4: AI Strategies for Robust Automotive AR - Develop the vision-based engine component identification

Appendix E: Ethical Documents

The project adheres to strict ethical guidelines regarding data collection and AI evaluation.

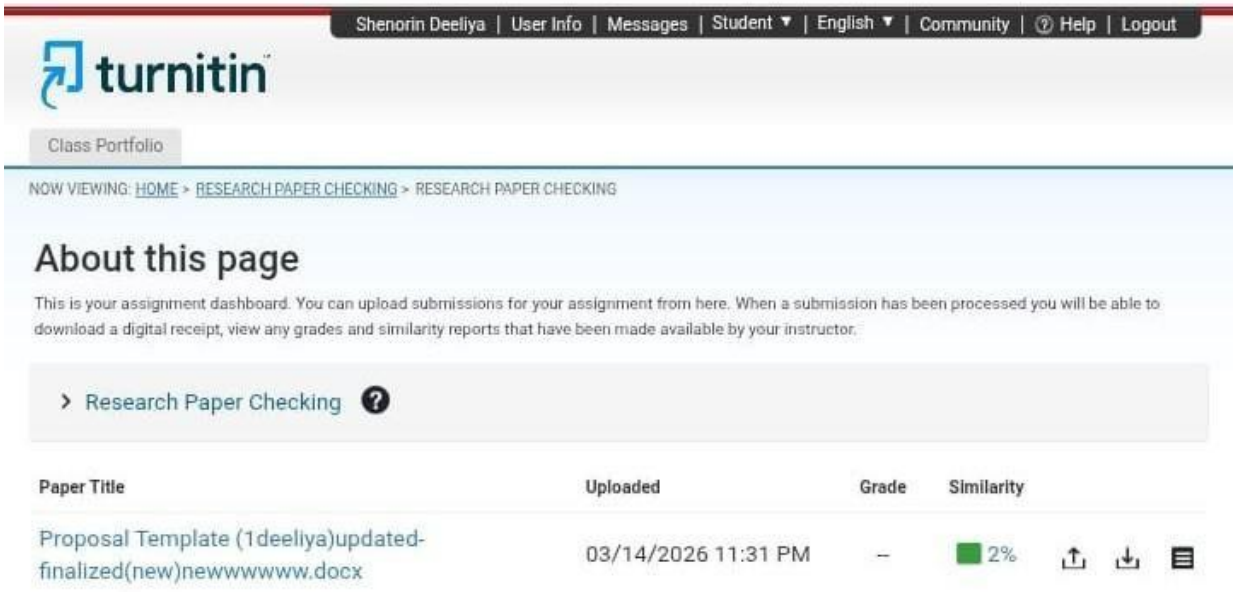
Ethical Clearance: Explicit clearance is required if collecting identifiable student video data.

Data Integrity: Dataset collection primarily utilizes instructor reference videos and selfrecorded footage of Alto 800L engine to ensure privacy.

On-Device Processing: All vision processing (component detection + fingertip tracking) occurs on-device. No video data is sent to external servers.

Unbiased Evaluation: The system provides automated, unbiased AI-based scoring to ensure fair skill assessment.

SDG Alignment: Documented commitment to SDG 4 (Quality Education) and SDG 12 (Responsible Consumption) by reducing component waste during training on vehicles like the Alto 800L.


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